

Week 1 Task: Data Acquisition, Cleaning & EDA Report

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Dataset: Titanic Passenger Cleaned Dataset (891 records)

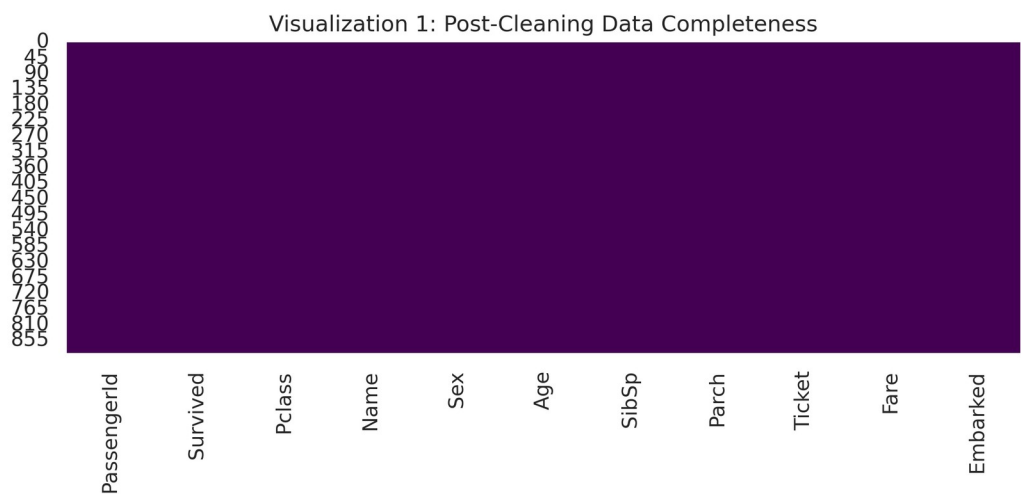
1. Executive Summary & Data Cleaning Rationale

The primary objective of Week 1 was to execute an end-to-end data preparation and exploratory analysis workflow. The following cleaning procedures were implemented:

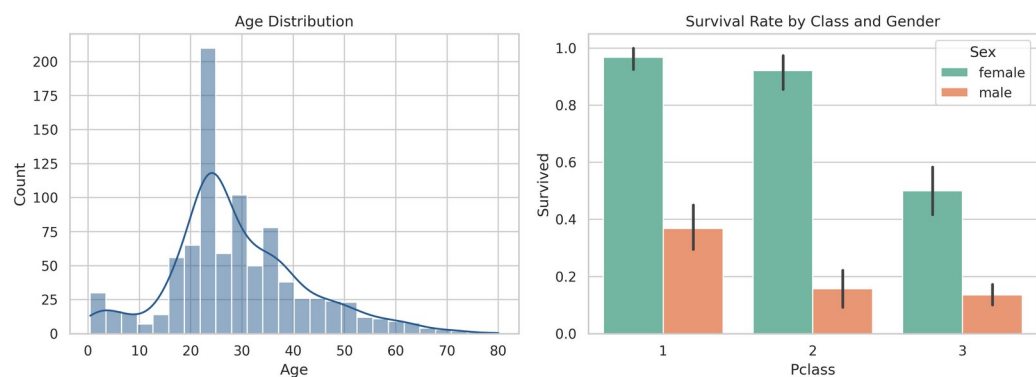
- Missing Age Handling: Imputed missing age records using the median age grouped by passenger class (Pclass) to preserve socio-economic stratification.
- Missing Embarked Handling: Imputed missing categorical values using the mode ("S").
- Column Dropping: Dropped the "Cabin" column due to an excessive proportion (>77%) of missing entries.
- Deduplication: Verified zero duplicate rows present in the raw dataset.

2. Exploratory Data Analysis & Visual Outputs

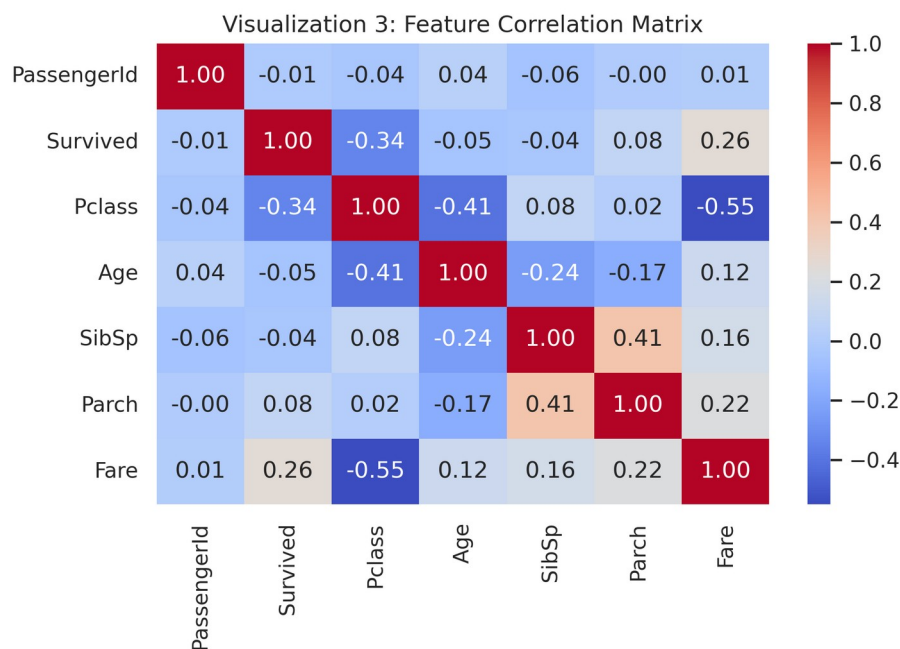
Data Completeness Check



Demographic Trends vs. Survival



Numerical Feature Correlations



3. Key Analytical Insights

- 1. Demographic Disparity: Females demonstrated a drastically higher survival rate across all passenger classes compared to males.
- 2. Socio-Economic Class Effect: First-class passengers (Pclass 1) achieved significantly higher survival odds than third-class passengers (Pclass 3).
- 3. Fare Correlation: Ticket fare shows a positive correlation (+0.26) with survival outcome, corroborating the Pclass trend.